

Alexandre Binninger

✉ alexandre.binninger@inf.ethz.ch

🌐 alexandrebinninger.com

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Work Experience

- 2026 – current **BRIDGE PoC Fellow (ETH Zurich)**, Founder of Boldbobbin
Building a modern embroidery design platform.
Leading R&D on automated digitization.
- 2021 – 2026 **Interactive Geometry Lab (ETH Zurich)**, Scientific assistant / Doctoral Student
Thesis title: Shape representations for intuitive modeling and generation.
Supervised by Prof. Dr. Olga Sorkine-Hornung.
- 2023 (4 months) **Meta**, Research scientist internship
Integrated the Gemini research group to work on 3D neural implicit representations.
Supervised by Dr. Philipp Herholz.
- 2019 (6 months) **Programming Methodology Group (ETH Zurich)**, Research intern
Study of the Completeness, Termination and Efficiency for e-matching-based Axiomatisations.
Supervised by Dr. Alexander J. Summers.
- 2018 (3 months) **Muvee Technologies**, Software engineering internship.
Integrated a development team working on an automatic photobook-maker application.
- 2016 – 2017 **Gendarmerie Nationale**, Leadership training as Cadet Officer
École Polytechnique's first-year military service. Optimized duty hours across police units.

Education

- 2021 – 2025 **Ph.D. in Computer Science, ETH Zurich** Computer Graphics
Research interests: shape representations, generative AI, shape modeling.
- 2019 – 2021 **M.Sc. Computer Science, ETH Zurich**.
Focus on visual computing courses.
Thesis: *Approximation of 3D Shapes by Developable Surfaces via Tangent Space Thinning*.
- 2016 – 2019 **Master of Engineering, Science and Technology, École Polytechnique**.
Algorithms and foundations of programming languages track.
- 2017 – 2018 **Bachelor of Philosophy, Université Paris-Nanterre**.
Distance learning alongside scientific studies. Awarded with honors.
- 2014 – 2016 **Preparatory Program** Lycée Louis-Le-Grand.
A two-year post secondary intensive coursework leading to competitive entrance examinations to the *Grandes Écoles*. Main subjects: Mathematics and Physics.

Research Publications

- 1 R. Wiersma*, **A. Binninger***, P. Li, T. Magne, A. Oehri, A. Segall, D. Luterbacher, M. Padilla, J. Ren, and O. Sorkine-Hornung, "Gradient descent in the ALPS: Abstracted low-poly stylization and fabrication," in *SIGGRAPH Conference Papers '26*, *Equal contribution, New York, NY, USA: ACM, 2026, ISBN: 979-8-4007-2554-8/2026/07.

- 2 **A. Binninger**, R. Wiersma, P. Herholz, and O. Sorkine-Hornung, “TetWeave: Isosurface Extraction using On-The-Fly Delaunay Tetrahedral Grids for Gradient-Based Mesh Optimization,” *ACM Trans. Graph.*, vol. 44, no. 4, Aug. 2025, *SIGGRAPH 2025 issue*.
- 3 **A. Binninger** and O. Sorkine-Hornung, “SD- π XL: Generating Low-Resolution Quantized Imagery via Score Distillation,” in *SIGGRAPH Asia 2024 Conference Papers*, ser. SA '24, Tokyo, Japan: Association for Computing Machinery, Dec. 2024, ISBN: 9798400711312.
- 4 **A. Binninger**, A. Hertz, O. Sorkine-Hornung, D. Cohen-Or, and R. Giryes, “SENS: Part-Aware Sketch-based Implicit Neural Shape Modeling,” *Computer Graphics Forum*, vol. 43, no. 2, e15015, Apr. 2024, *Honorable Mention*.
- 5 **A. Binninger** and O. Sorkine-Hornung, “Smooth Interpolating Curves with Local Control and Monotone Alternating Curvature,” *Computer Graphics Forum (proceedings of SGP 2022)*, vol. 41, no. 5, Jul. 2022, ISSN: 1467-8659.
- 6 **A. Binninger***, F. Verhoeven*, P. Herholz, and O. Sorkine-Hornung, “Developable Approximation via Gauss Image Thinning,” *Computer Graphics Forum (proceedings of SGP 2021)*, vol. 40, no. 5, pp. 289–300, Jul. 2021, **Equal contribution*, ISSN: 1467-8659.

Skills

Languages	 French (native), English (proficient), German (intermediate), Mandarin Chinese (intermediate).
Programming	 Python, TypeScript, C++, Rust. Notions in Java, Lua, C, Bash.
AI & Graphics Libraries	 PyTorch, libigl, OpenGL, ThreeJS, wgpu.
Web Development	 HTML, CSS, Node.js, React, Tailwind CSS, MySQL.
Design & Visualization	 Blender, Adobe Illustrator, Inkscape, Adobe Premiere Pro.
Technical Tools	 $\LaTeX$, Git, Unix, Docker.
Research & Professional	 Academic research, scientific writing, project management, teaching, public speaking.

Outside Activities

Miscellaneous Experience

- 2024 – 2025  **Talent Kick**, TK-S24 batch
 Talent Kick is an entrepreneurship program for students enrolled in Swiss Universities. Founded Fabricaible, a start-up providing generative models to create fabricable designs. Pitched at venture pitching event and in front of the Talent Kick strategy board.
- 2023 – 2025  **VMI board member**, ETH D-INFK scientific staff association
 VMI represents, supports and connects the scientific staff of the computer science department of ETH.
 Organized many events, such as 3 career-focused panel discussions with a focus on academic excellence.
 Co-president between Oct. 2024 and Feb. 2025.
- 2021 – 2025  **Teaching Assistant**, ETH Zurich
 Taught 2 semesters in Linear Algebra, 3 semesters in Shape Modeling courses.
 One semester as Head TA for Linear Algebra. Managed other teaching assistants, created graded homework, and designed the final exam.

Outside Activities (continued)

- 2019 – 2020  **Discord community - Developer and Leader**, E-sport Community
Developed a Discord server reaching 1'000 members. Tournament Organizer.
Designed and implemented a Discord bot hosted on a remote VPS featuring a ranking system, a match wizard, an automated competitive event manager and a profile card generator with gamified features.
Implementation with discord.py, Python Imaging Library (PIL) and MySQL.
- 2017 – 2018  **Team Research Project**, Live theater subtitling Project
Displayed subtitles in real time on smart glasses or a smartphone with a web interface.
Designed the algorithm matching the actors' spoken text with their written text.
Winners of the Vivatech Competition. Project presented at the international Microsoft Imagine Cup.
-  **Binet Réseau**, Computer Technology association at École Polytechnique
Deployed a web service for equipment pool among students on the campus.
Integrated the Sysadmin team. Provided support for students' computer-related problems.

Hobbies

- Sports  Calisthenics, running, hiking, volleyball.
- Cultural  Philosophy, self-help books, video games, cooking.
- Creative  Embroidery machine.